k-Nearest Neighbors (algorithm) Logistic Regression Regression (statistics)

Classification (machine learning) Statistics (academic discipline) Machine Learning

How can we compare the advantages and disadvantage of logistic regression versus k-nearest neighbour?

1 Answer

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Answered Sep 27, 2015

Here are some points of comparison:

- **Training**: k-nearest neighbors requires no training. Logistic regression requires some training. On the other hand, in k-nearest neighbors, you need to tune k. Logistic regression doesn't need any parameter tuning.
- **Decision boundary**: Logistic regression learns a linear classifier, while k-nearest neighbors can learn non-linear boundaries as well.
- **Predicted values**: Logistic regression predicts probabilities, which are a measure of the confidence of prediction. k-nearest neighbors predicts just the labels.

However, none of these comparisons tell you which method will perform better in general -- they are fundamentally different, and one can find examples where one is better than the other.

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